Major Project 21CSA699A

Project_Proposal

- **Title:** AI-Based Smart Attendance System using Face Recognition
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- **Abstract:**
This project aims to develop an automated attendance system using face recognition technology. Traditional attendance methods are time-consuming and prone to errors such as proxy attendance. The proposed system uses a camera to capture real-time images, detects faces, and compares them with a pre-trained dataset to identify individuals. Once recognized, attendance is automatically recorded with date and time.
The system utilizes computer vision and machine learning techniques to improve accuracy and efficiency. This project reduces manual effort, ensures reliability, and enhances security in attendance management systems.

- **Assumptions/Declarations:**

- The system assumes that student images are pre-registered in the database.
- Proper lighting conditions are required for accurate face detection.
- Each student's face is unique and can be identified reliably.
- The system will be used in a controlled environment such as a classroom.

- **Main Objective/Deliverable:**

- To design and implement a **face recognition-based attendance system**
- To automatically detect and recognize faces using a webcam
- To record attendance with timestamp in a database/CSV file
- To reduce manual errors and prevent proxy attendance

- **Timeline and Milestones:**

Milestone	Timeline
Proposal Submission	Week 2
Dataset Collection & Preprocessing	Week 3-4
Model Development (Face Encoding)	Week 5-6
Attendance System Implementation	Week 7-8
Testing and Validation	Week 9
UI Development & Improvements	Week 10-12
Final Report Preparation	Week 13-14

- **Tools to be used:**

Software/Hardware Tools	Specifications
Python	Programming language for implementation
OpenCV	Image processing and face detection
face_recognition (dlib)	Face encoding and recognition
NumPy	Numerical computations
Pandas	Data handling and CSV storage
Laptop/Desktop	Minimum 8GB RAM, i5 processor recommended
Webcam	Built-in or external camera for real-time capture

- **Learning involved:**

- Understanding of **Computer Vision concepts**
- Face Detection and Recognition techniques
- Image processing using OpenCV
- Machine Learning basics (feature extraction, matching)
- Real-time system development
- Data storage and handling using CSV/Database
- Building simple UI for user interaction

Date : 02-05-2026	
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